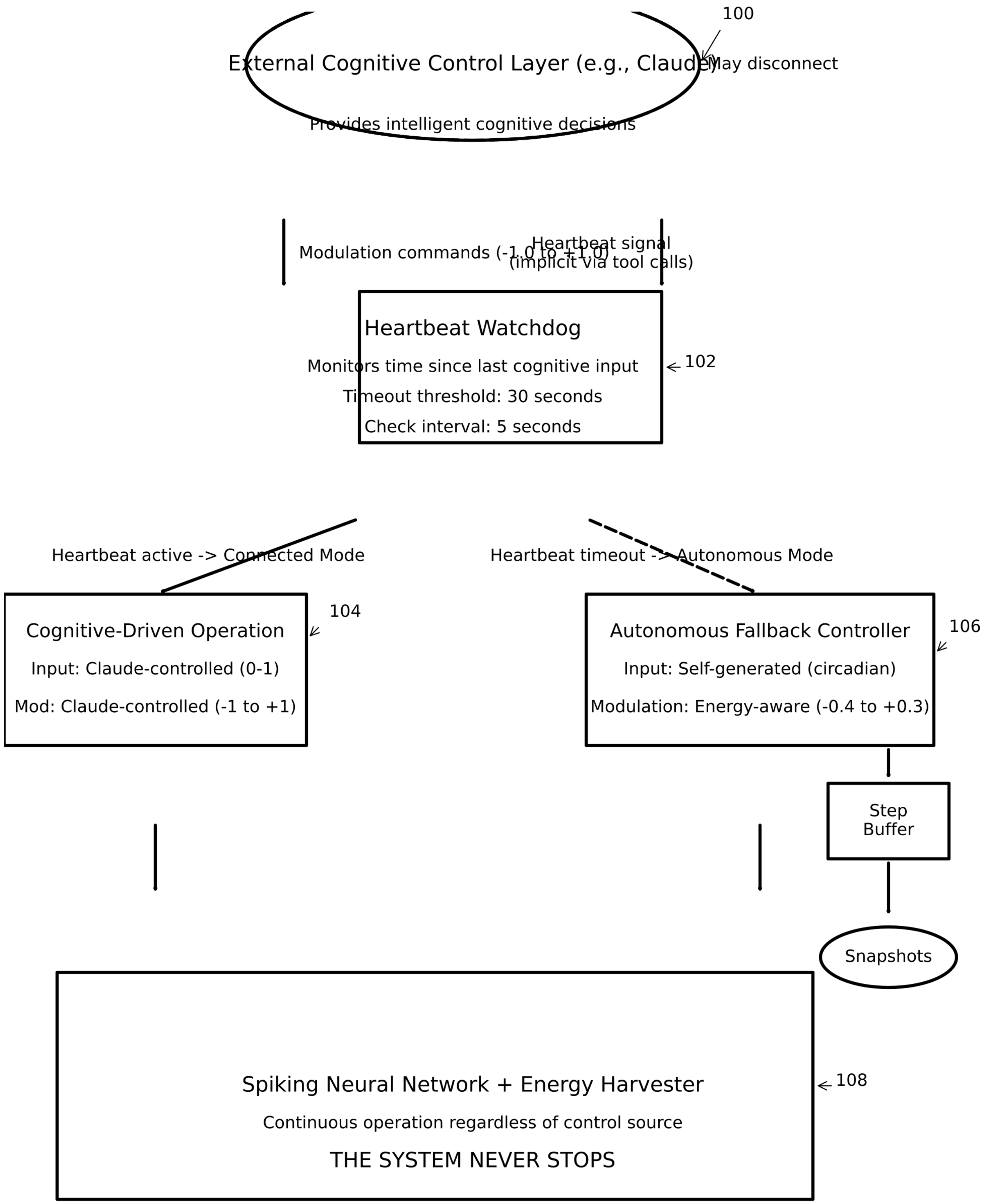


Seamless transition: SNN continues processing without interruption

The control source changes; the computation continues

FIG. 1

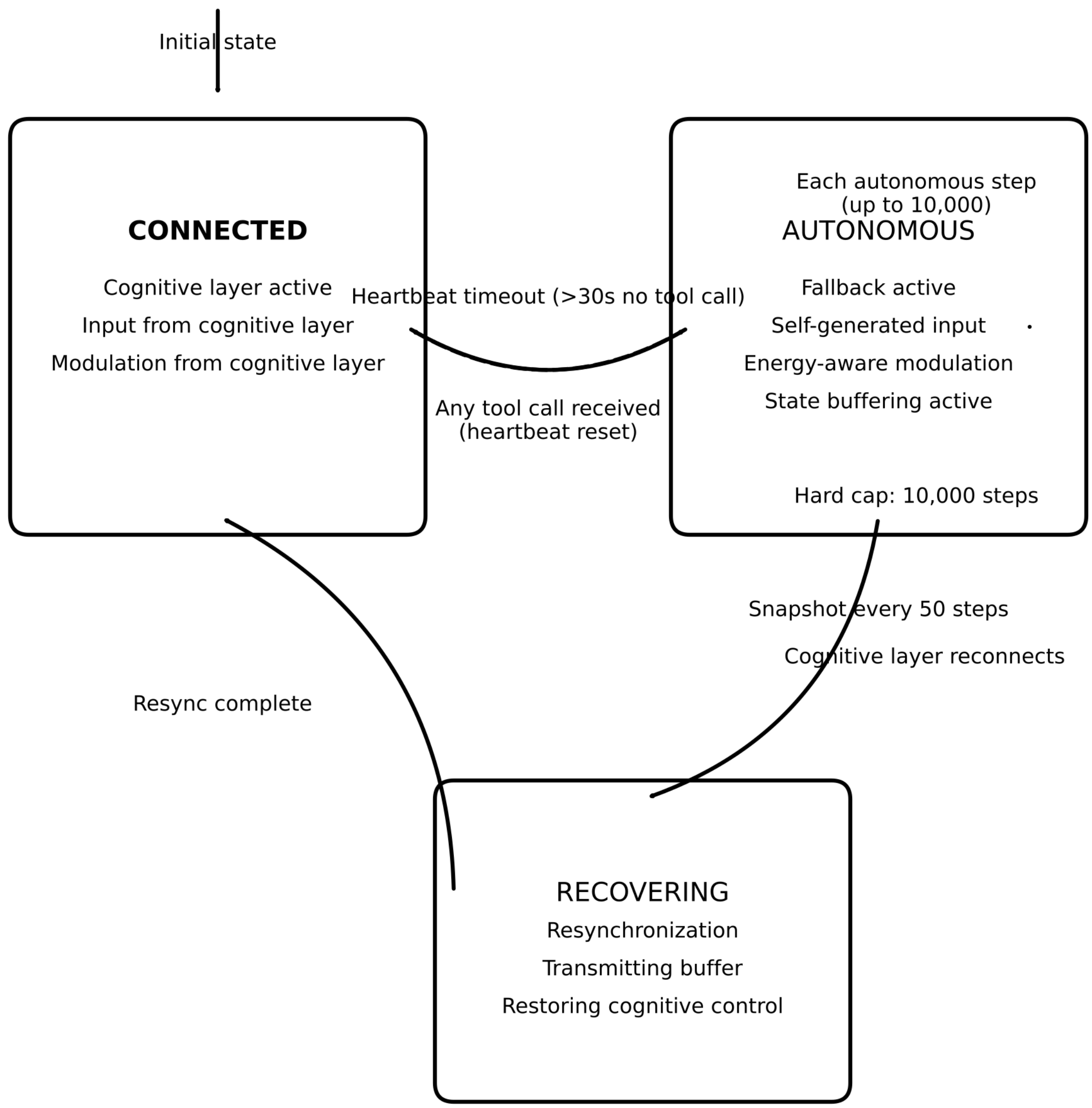
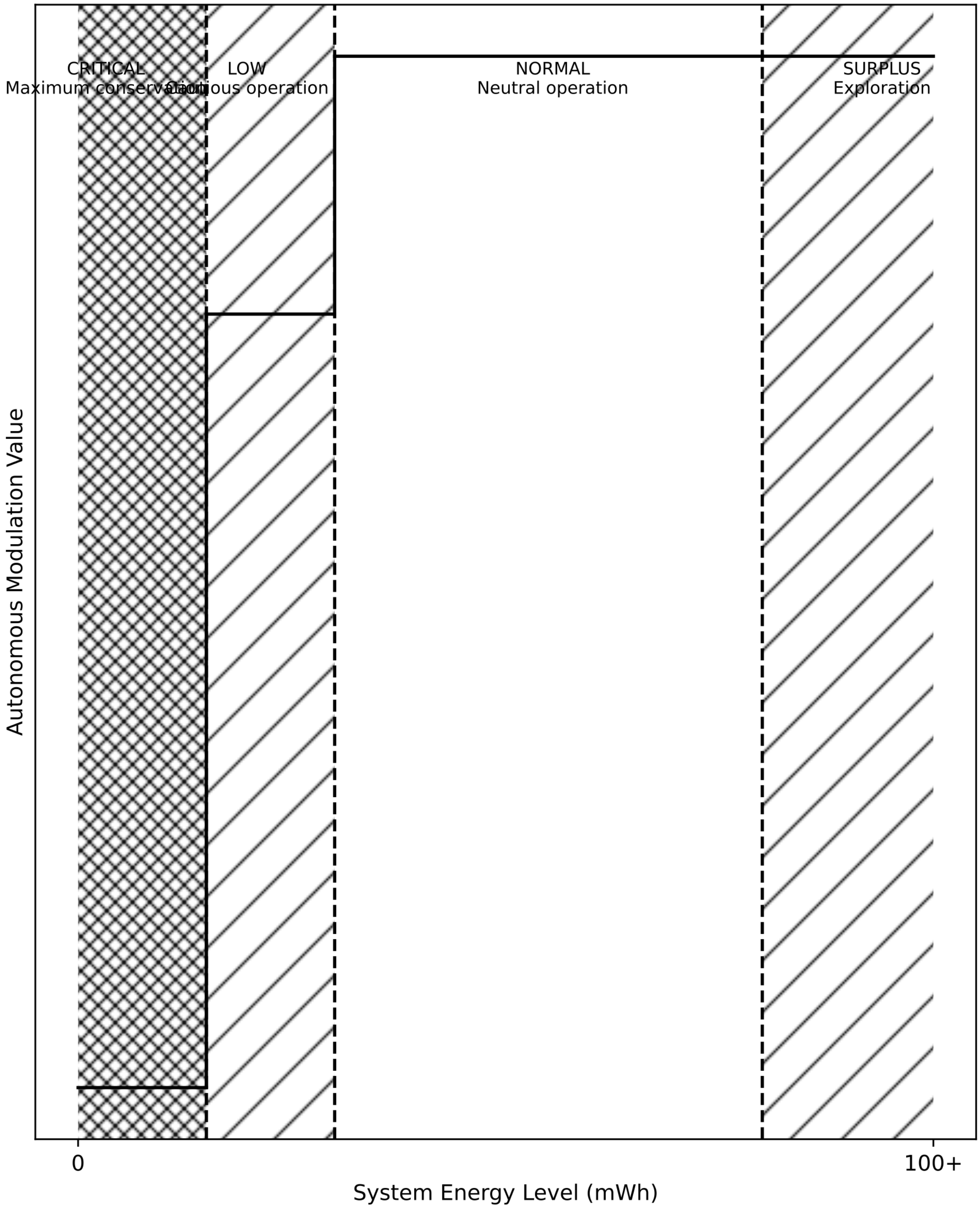
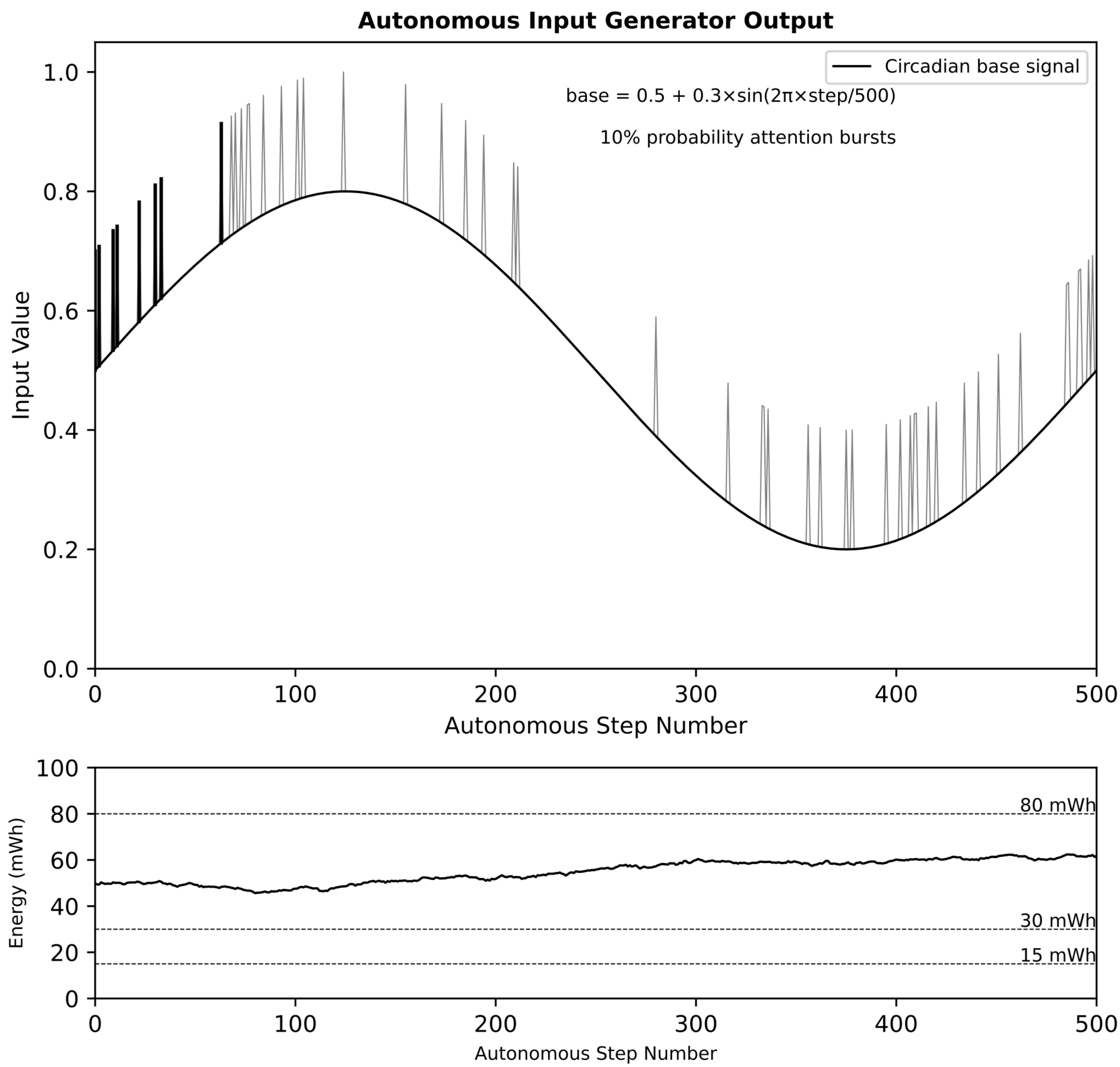

Initial state
CONNECTED
Cognitive layer active
Input from cognitive layer
Modulation from cognitive layer
Heartbeat timeout (>30s no tool call)
Any tool call received
(heartbeat reset)
Each autonomous step
(up to 10,000)
AUTONOMOUS
Fallback active
Self-generated input
Energy-aware modulation
State buffering active
Hard cap: 10,000 steps
Snapshot every 50 steps
Cognitive layer reconnects
Resync complete
RECOVERING
Resynchronization
Transmitting buffer
Restoring cognitive control

FIG. 2



System autonomously adjusts processing intensity based on energy — mimicking metabolic regulation

FIG. 3



Energy-aware modulation adjusts intensity

FIG. 4

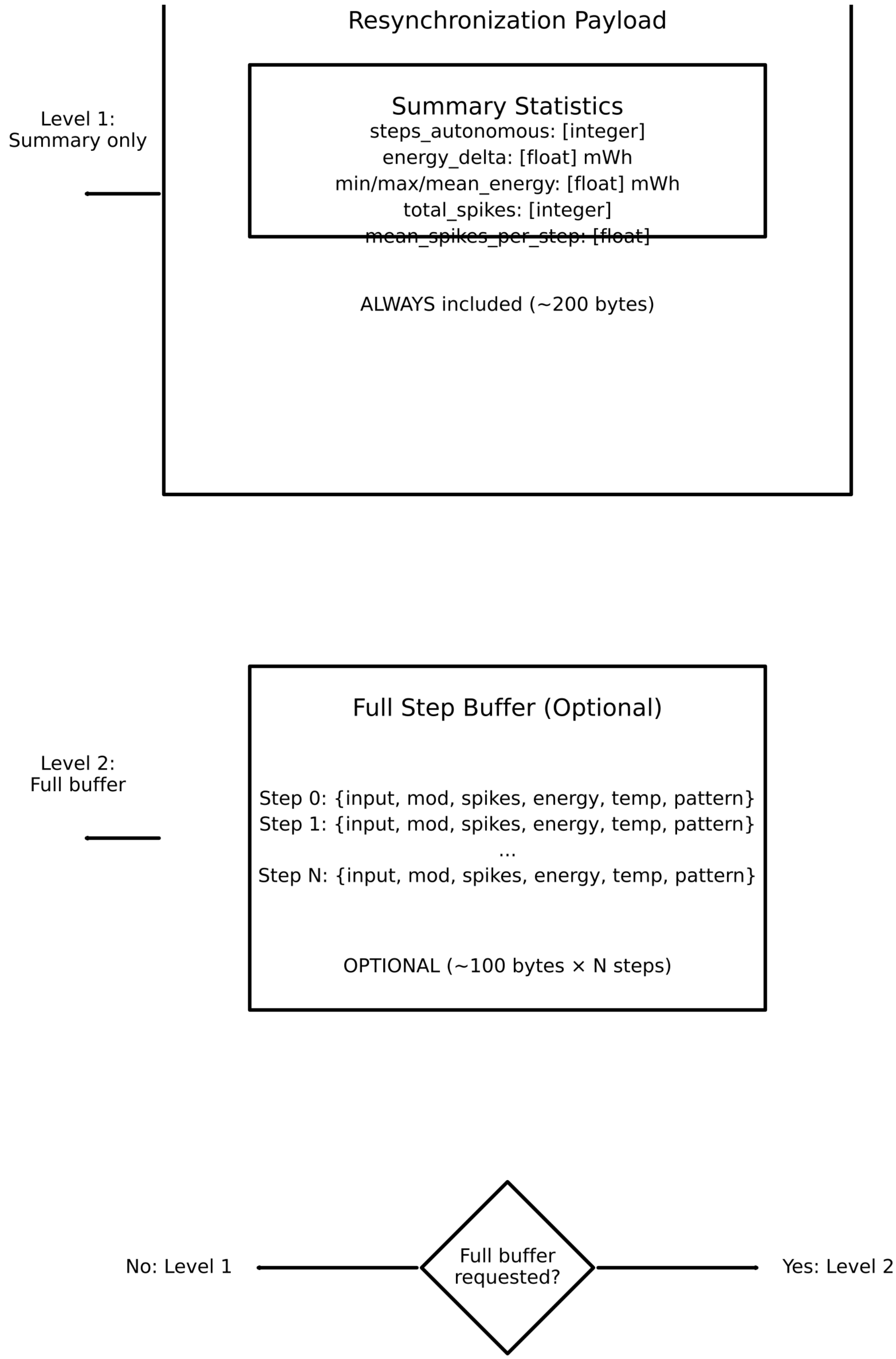

Resynchronization Payload
Level 1:
Summary only
Summary Statistics
steps_autonomous: [integer]
energy_delta: [float] mWh
min/max/mean_energy: [float] mWh
total_spikes: [integer]
mean_spikes_per_step: [float]
ALWAYS included (~200 bytes)
Level 2:
Full buffer
Full Step Buffer (Optional)
Step 0: {input, mod, spikes, energy, temp, pattern}
Step 1: {input, mod, spikes, energy, temp, pattern}
...
Step N: {input, mod, spikes, energy, temp, pattern}
OPTIONAL (~100 bytes × N steps)
No: Level 1
Full buffer requested?
Yes: Level 2

FIG. 5

Timeline A — Normal Reconnection:

Connected	Timeout	Autonomous	Resync	Connected
[Claude active]	[Watchdog]	[Self-regulated]	[Buffer sent]	[Restored]

Timeline B — Crash Recovery:

Connected	CRASH	Restart	Snapshot	Connected
[Normal op]	[Server dies]	[Relaunched]	[latest.json]	[Restored]

Max state loss = 50 steps (snapshot interval)

Timeline C — Clean Shutdown:

Connected	Autonomous	EOF Signal	Shutdown
[Normal op]	[Fallback]	[stdin closed]	[Snapshot written]

State preserved for next session — zero data loss

Solid segments: System actively processing

Dashed segments: System in transition

Bold marker at each state change: No data loss at any transition

FIG. 6

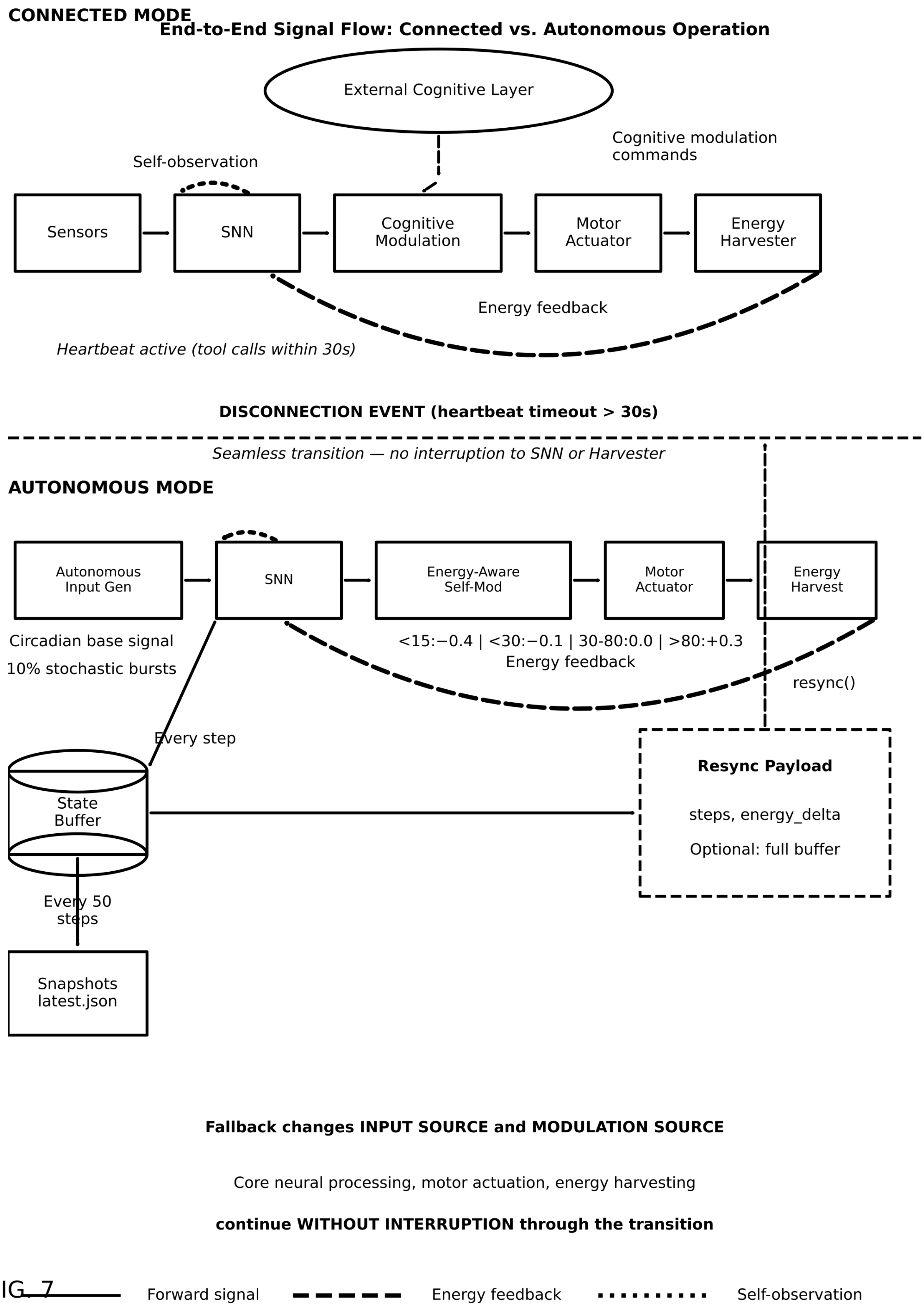

CONNECTED MODE
End-to-End Signal Flow: Connected vs. Autonomous Operation
External Cognitive Layer
Cognitive modulation commands
Self-observation
Sensors
SNN
Cognitive Modulation
Motor Actuator
Energy Harvester
Energy feedback
Heartbeat active (tool calls within 30s)
DISCONNECTION EVENT (heartbeat timeout > 30s)
Seamless transition — no interruption to SNN or Harvester
AUTONOMOUS MODE
Autonomous Input Gen
SNN
Energy-Aware Self-Mod
Motor Actuator
Energy Harvest
Circadian base signal
10% stochastic bursts
<15:−0.4 | <30:−0.1 | 30-80:0.0 | >80:+0.3
Energy feedback
resync()
Every step
State Buffer
Resync Payload
steps, energy_delta
Optional: full buffer
Every 50 steps
Snapshots latest.json
Fallback changes INPUT SOURCE and MODULATION SOURCE
Core neural processing, motor actuation, energy harvesting
continue WITHOUT INTERRUPTION through the transition
Forward signal
Energy feedback
Self-observation

FIG. 7